

Problem 1: SQL Querying

本题回忆细节有误，变得更难了（

```
1 CREATE TABLE train (  
2     trnid INTEGER PRIMARY KEY,  
3     name VARCHAR(50),  
4     type INTEGER  
5 );  
6  
7 CREATE TABLE station (  
8     staid INTEGER PRIMARY KEY,  
9     name VARCHAR(100),  
10    city VARCHAR(50),  
11    stop_sequence INTEGER,  
12    rank INTEGER  
13 );  
14  
15 CREATE TABLE timetable (  
16     id INTEGER PRIMARY KEY,  
17     train INTEGER,  
18     station INTEGER,  
19     arrival_time TIME,  
20     departure_time TIME,  
21     FOREIGN KEY (train) REFERENCES train(trnid),  
22     FOREIGN KEY (station) REFERENCES station(staid)  
23 );  
24  
25 CREATE TABLE ticket (  
26     tid INTEGER PRIMARY KEY,  
27     train INTEGER,  
28     src INTEGER,  
29     des INTEGER,  
30     price DECIMAL(10,2),  
31     date DATE,  
32     FOREIGN KEY (train) REFERENCES train(trnid),  
33     FOREIGN KEY (src) REFERENCES station(staid),  
34     FOREIGN KEY (des) REFERENCES station(staid)  
35 );
```

1. True or False: Which of the following SQL queries correctly finds the name with the most stops in the station table?

主要考察查询最大值的多种方法

```

1 SELECT s.name, COUNT(t.station) AS stop_count
2 FROM station s
3 JOIN timetable t ON s.staid = t.station
4 ORDER BY stop_count DESC
5 LIMIT 1;

```

```

1 SELECT s.name, COUNT(t.station) AS stop_count
2 FROM station s
3 JOIN timetable t ON s.staid = t.station
4 GROUP BY s.staid, s.name
5 ORDER BY stop_count DESC
6 LIMIT 1;

```

```

1 SELECT s.name, COUNT(t.station) AS stop_count
2 FROM station s
3 JOIN timetable t ON s.staid = t.station
4 GROUP BY s.staid, s.name
5 HAVING COUNT(t.station) = (
6     SELECT MAX(cnt)
7     FROM (
8         SELECT COUNT(station) AS cnt
9         FROM timetable
10        GROUP BY station
11     ) counts
12 );

```

```

1 SELECT s.name, COUNT(t.station) AS stop_count
2 FROM station s
3 JOIN timetable t ON s.staid = t.station
4 GROUP BY s.staid, s.name
5 HAVING COUNT(t.station) >= all (
6     SELECT COUNT(station) AS cnt
7     FROM timetable
8     GROUP BY station
9 );

```

2. True or False: Which of the following SQL queries correctly finds the cheapest price among tickets with origin A and destination B?

主要考察表连接的多种方法

```

1 SELECT t.tid
2 FROM ticket t
3 JOIN station src ON t.src = src.staid
4 JOIN station des ON t.des = des.staid
5 WHERE src.name = 'A' AND des.name = 'B'
6 ORDER BY t.price ASC
7 LIMIT 1;

```

```

1 SELECT t.tid
2 FROM ticket t, station src, station des
3 WHERE t.src = src.staid AND t.des = des.staid AND
4       src.name = 'A' AND des.name = 'B'
5 ORDER BY t.price ASC
6 LIMIT 1;

```

```

1 SELECT c.tid
2 FROM (
3     SELECT t.tid, t.price
4     FROM ticket t, station src, station des
5     WHERE t.src = src.staid AND t.des = des.staid AND
6           src.name = 'A' AND des.name = 'B'
7 ) c
8 ORDER BY c.price ASC
9 LIMIT 1;

```

```

1 SELECT t.tid
2 FROM ticket t
3 WHERE EXISTS (
4     SELECT 1
5     FROM station src
6     WHERE t.src = src.staid AND src.name = 'A'
7 )
8 AND EXISTS (
9     SELECT 1
10    FROM station des
11    WHERE t.des = des.staid AND des.name = 'B'
12 )
13 ORDER BY t.price ASC
14 LIMIT 1;

```

3. Fill in the blank to implement the query: Find the name of trains that depart later than 17:00:00.

```

1 SELECT ____ FROM train
2 WHERE EXISTS (
3     SELECT 1
4     FROM timetable
5     WHERE _____ AND
6           departure_time > '17:00:00'
7 );

```

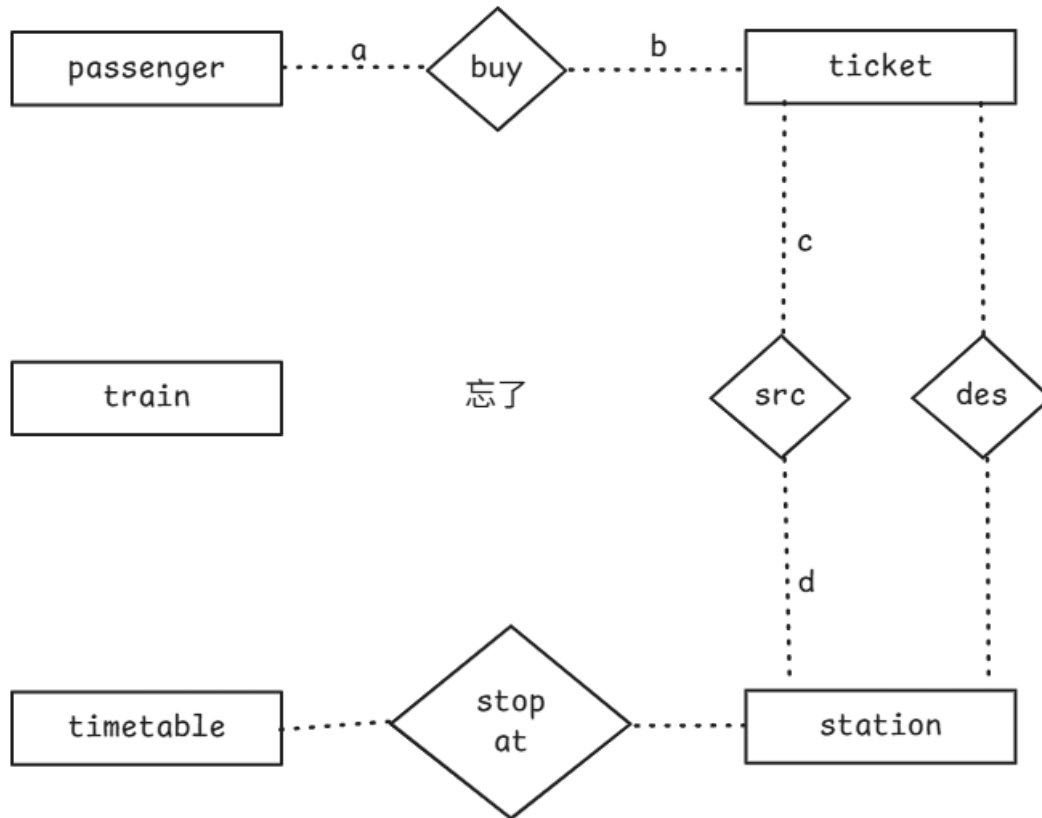
4. True or False: When joining the timetable and station tables, the condition `city = "Hangzhou"` can be applied to station first before joining with timetable.

5. True or False: When joining the timetable, station, and train tables, the conditions `city = "Hangzhou"` and `train.name = "ABC"` can be applied to station and train first before joining with timetable.

6. Write the relational algebra expression for the following query: Find the name and type of trains that stop at the "Hangzhou Dong" station:

Problem 2: E-R Model

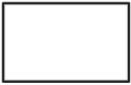
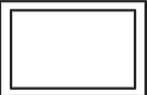
本题回忆细节有误



Assumption:

- A passenger can purchase multiple tickets.
- A ticket may remain unsold.
- A station must serve as the source (src) for at least one ticket.

1. If timetable is a weak entity set, with train as its identifying entity set, how is the relationship set connecting them represented?

A.  B.  C.  D. 

2. What is the shape of edge a?

A. undirected line B. undirected double line C. directed line D. directed double line

3. What is the shape of edge b?

A. undirected line B. undirected double line C. directed line D. directed double line

4. What is the shape of edge c?

A. undirected line B. undirected double line C. directed line D. directed double line

5. What is the shape of edge d?

- A. undirected line B. undirected double line C. directed line D. directed double line

Problem 3: Relational Formalization

The relation $R = \{A, B, C, D, E, F\}$ has the following functional dependencies (FDs):
 $F = \{A \rightarrow B, AC \rightarrow D, BC \rightarrow D, E \rightarrow F, AEF \rightarrow C\}$.

1. True or False: The closures $\{A, C, F\}^+$, $\{A, B, C, F\}^+$, $\{A, C, D, F\}^+$, and $\{A, B, C, D, F\}^+$ contain the same attributes.
2. The canonical cover of F is: _____
3. Which of the following is a candidate key of R ?
A. AEF B. AE C. AC D. ACE E. None of the above
4. Adding which of the following dependencies will **NOT** change the candidate key(s)?
A. $A \rightarrow E$ B. $E \rightarrow A$ C. $B \rightarrow C, C \rightarrow E$ D. $D \rightarrow CE$ E. None of the above
5. True or False: Decomposing R into (A, B, E, F) and (A, B, C, D, F) is **NOT** lossless.
6. True or False: Decomposing R into (A, B) , (A, C, D) , and (A, C, E, F) results in all three subrelations satisfying BCNF.

Problem 4: Storage and File Structure

本题回忆细节有误

1. True or False:

Cache is primary storage and main memory is secondary storage.

DBMS transfers data from disk to main memory.

Data on the inner tracks of a disk is accessed slower than data on the outer tracks.

A 15,000 RPM disk does not necessarily have a shorter seek time than a 7,500 RPM disk.

```
1 CREATE TABLE A (  
2     id INTEGER PRIMARY KEY,  
3     name VARCHAR,  
4     type INTEGER  
5 )
```

Assumption:

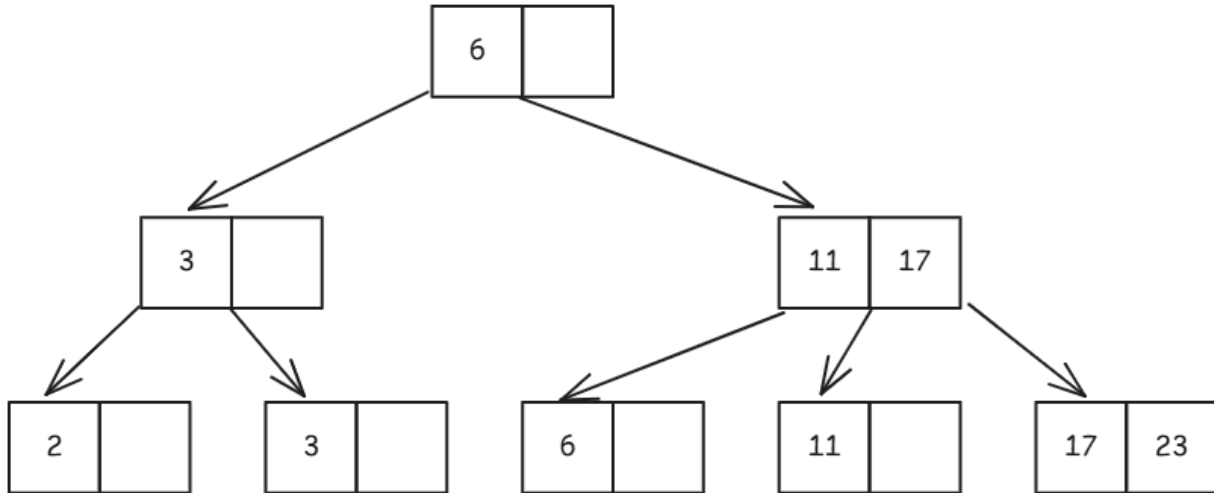
- Integer size: 4B
 - `name` field's offset and length each occupy 4B
 - Null bitmap is omitted
2. Given a record `(23, "chicago", 10)`, fill in its memory structure:

23												C	h	i	c	a	g	o
0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18

3. If NULL values are allowed, the minimum size of a record is: ____ B
4. Using a slotted-page structure, where: Free space pointer occupies 4B; Record count field occupies 4B; Length and pointer for each record occupy 2B each. The maximum number of records that can be stored in a 4KB page is: ____

Problem 5: Indexing

Given the following B+ Tree with $n = 3$:



1. After inserting 27, the root node contains:
A. 6, 17 B. 6 C. 11 D. 11, 17 E. None of the above
2. After insertion, if we then delete 11, the root node contains:
A. 6 B. 17 C. 6 or 17 D. None of the above

```

1 CREATE TABLE T (
2     id INTEGER PRIMARY KEY,
3     a VARCHAR(50),
4     b INTEGER
5 );
  
```

The table T has a B+ Tree index on `id` with $n = 100$, and there are 300,000 records in total.

3. The maximum height of the B+ Tree is: ____
4. The maximum data transfer required to search for a specific record by `id` is: ____

Problem 6: Query Processing

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1. True or False:

Linear search is applicable in all cases

Index search is always faster than linear search

In external sorting, relation r has 400 records, relation s has 3000 records, $M = 20$, $b_b = 1$

2. The initial number of generate runs for r and s are ___ and ___ respectively
3. True or False: Both r and s require exactly 2 passes during external sorting
4. After external sorting, when using merge join. The data transfer cost is ____, The seek cost is ____
5. True or False:
 If $M > 3$, $M-2$ blocks should be allocated for the outer relation
 If $b_b = 20$, then allocating 1 block to r, 18 blocks to s, and 1 block to output would minimize the join cost

Problem 7: Query Optimization

本题回忆细节有误

1. The following equivalence relations are incorrect:

- A. $\sigma_{\theta}(E_1 - E_2) = \sigma_{\theta}(E_1) - \sigma_{\theta}(E_2)$
- B. $\sigma_{\theta}(E_1 - E_2) = \sigma_{\theta}(E_1) - E_2$
- C. $\sigma_{\theta}(E_1 \cap E_2) = \sigma_{\theta}(E_1) \cap E_2$
- D. $\sigma_{\theta}(E_1 \cup E_2) = \sigma_{\theta}(E_1) \cup E_2$

```

1 CREATE TABLE student (
2     sid INTEGER PRIMARY KEY,
3     name VARCHAR(20) NOT NULL,
4     gender CHAR(1) CHECK (gender IN ('M', 'F')),
5 );
6
7 CREATE TABLE grade (
8     sid INTEGER PRIMARY KEY,
9     grade INTEGER,
10    FOREIGN KEY (sid) REFERENCES student(sid)
11 );
  
```

```

1 select s.sid
2 from student s, grade g
3 where s.sid = g.sid AND
4       grade > 107;
  
```

Assumption:

- `sid` has 50 distinct values
 - `grade` ranges from 60 to 120 (inclusive), uniformly distributed
2. The selectivity of condition `s.sid = g.sid` is: ____
 3. he selectivity of condition `grade > 107` is: ____

Problem 8: Transaction and Concurrency Control

1. True or False

- A conflict-serializable schedule must be a recoverable schedule.
- Conflict serializability corresponds to a unique serial schedule.
- 2PL (Two-Phase Locking) ensures recoverability.
- Strict 2PL ensures recoverability.

T1	T2	T3
	Read(A)	
	Write(A)	
Write(A)		
Read(B)		
		Read(B)
		Write(B)
	Read(B)	
Read(C)		
	Write(C)	

2. Draw the precedence graph. What are the edges?

- A. $\{T1 \rightarrow T2, T3 \rightarrow T1, T2 \rightarrow T1, T1 \rightarrow T3, T3 \rightarrow T2\}$
- B. $\{T1 \rightarrow T2, T2 \rightarrow T1, T1 \rightarrow T3, T3 \rightarrow T2, T2 \rightarrow T1\}$
- C. $\{T1 \rightarrow T2, T2 \rightarrow T1, T1 \rightarrow T3, T3 \rightarrow T2\}$
- D. None of above

原题 B 选项有问题，本文档保持原内容不变

3. Is this schedule conflict-serializable?

4. True or False: Deadlock

- Deadlock can be represented in a precedence graph.
- The Tree Protocol can prevent deadlocks.
- Rigorous 2PL can prevent deadlocks.
- The Wait-Die scheme can prevent deadlocks.

Problem 9: Recovery


```

1 3000: <T1, 6421.1, 10, 20>
2 3001: <T2, start>
3 3002: <T3, start>
4 3003: <T2, 3462.1, 30, 40>
5 3004: <T3, 7923.1, 50, 60>
6 3005: <T2, 3462.1, 40, 70>
7 3006: <T1, 6421.1, 20, 80>
8 3007: <T3, commit>
9 3008: checkpoint
10
11      DirtyPageTable
12      Page PageLSN RecLSN
13      3462 3005    3005
14      6421 3006    3000
15      7923 3004    3004
16
17      TXN LastLSN
18      T1   3006
19      T2   3005
20
21 3009: <T4, start>
22 3010: <T4, 5235.1, 60, 90>
23 3011: <T1, 6421.1, 80, 100>
24 3012: <T1, commit>

```

1. The Analysis phase starts at:

A. 3000 B. 3001 C. 3002 D. 3008 E. None of the above

2. After the Analysis phase completes, what are the DirtyPageTable and undo-list? Fill in the tables below (leave a row empty if an entry does not exist).

Page	PageLSN	RecLSN
3462		
5235		
6421		
7923		

TXN	LastLSN

3. The Redo phase starts at:

A. 3000 B. 3001 C. 3002 D. 3008 E. None of the above

4. During the Redo phase, which log entries are skipped?

A. 3000 B. 3003 C. 3010 D. 3011 E. None of the above

5. The first log entry to be undone is:

A. 3011 B. 3010 C. 3006 D. 3005 E. None of the above

6. The LSN of the CLR (Compensation Log Record) for log entry 3005 is:

A. 3013 B. 3014 C. 3015 D. 3016 E. None of the above

7. After recovery completes, what are the values of 3462.1 and 6421.1?

A. 70, 10 B. 30, 10 C. 30, 100 D. 70, 100 E. None of the above